

Deep inelastic scattering on polarized lithium: the physics behind the planned measurements

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Polarized lithium beams would provide the EIC with spin-polarized *nuclei* light enough for ab initio computation yet complex enough to exhibit genuine nuclear structure. ${}^6\text{Li}$ is a spin-1 nucleus well described as a deuteron bound to an inert α core; ${}^7\text{Li}$ is its spin-3/2 partner, an α plus a triton. Starting from the DIS formalism, this note develops the four measurements the program is built around: the tensor structure function b_1 , the double-helicity-flip ("exotic glue") structure function Δ , the polarized EMC effect, and spectator-tagged DIS on the embedded deuteron.

1 · Why polarized lithium, and why at a collider

The EIC project scope provides polarized electrons, protons and ${}^3\text{He}$; polarized deuterons are planned as a later addition, and no heavier polarized ion appears in the conceptual design or the Yellow Report [29,25]. The EPIOS white paper [1] makes the case for polarized ${}^6\text{Li}$ and ${}^7\text{Li}$ — "a desirable addition", listed with ${}^2\text{H}$ and ${}^3\text{He}$ as critical technology — and the ANL source development [51] specifies the beam: an atomic-beam source with optical pumping at 671 nm feeding EBIS charge breeding, at source targets $P_z \geq 0.90$ (vector) and $P_{zz} \geq 0.80$ (tensor) and $\geq 10^{13}$ atoms/s. Four measurements motivate it, each addressing a gap in the literature: no b_1 prediction or measurement exists for any nucleus beyond the deuteron; no gluon-transversity projection exists for any EIC target; no study of α -tagged DIS on lithium exists; and no lithium-beam luminosity or polarization parameter set has been published. The program's simulations convert these gaps into figures of merit.

1.1 · Why lithium as a source: an alkali atom with the right nuclei

Lithium is the lightest alkali metal and its atom is hydrogen-like: a single 2s valence electron whose D lines near 671 nm (670.96 and 670.98 nm in vacuum for ${}^7\text{Li}$) lie within the range of commercial single-frequency diode lasers, and whose ground-state hyperfine splittings, 228.2 MHz for ${}^6\text{Li}$ and 803.5 MHz for ${}^7\text{Li}$ [35], are small enough for one laser with an electro-optic sideband to address both hyperfine levels. Circularly polarized light pumps the atoms into a single magnetic substate, the hyperfine coupling transfers the electron polarization to the nucleus, and radio-frequency transitions between Zeeman sublevels set any combination of P_z and P_{zz} [51]. Boron, the other light candidate with a large nuclear spin, "presents a challenge for optical pumping" [1]. The low ionization energy of lithium (5.39 eV [35]) eases ionization, and fully stripped Li^{3+} is within reach of the EBIS being commissioned to charge-breed the polarized ${}^3\text{He}^{++}$ beam, whose design goal is 2×10^{11} ions per pulse at $\geq 70\%$ polarization [34,1].

The nuclei are the other half of the argument. Both isotopes are stable and abundant (7.6% ${}^6\text{Li}$ and 92.4% ${}^7\text{Li}$ in natural lithium), and they are the only stable isotopes of one element with $J = 1$ and $J = 3/2$: the stable spin-1 nuclei are exactly ${}^2\text{H}$, ${}^6\text{Li}$ and ${}^{14}\text{N}$, the stable spin-3/2 nuclei below $A = 30$ are ${}^7\text{Li}$, ${}^9\text{Be}$, ${}^{11}\text{B}$, ${}^{21}\text{Ne}$ and ${}^{23}\text{Na}$ [36]. One source, one laser system and one injector therefore serve a spin-1 and a spin-3/2 program. The EPIOS list of where an exotic-glue discovery is most likely — "spin-1 ${}^6\text{Li}$ or ${}^{14}\text{N}$, or spin-3/2 ${}^7\text{Li}$ or ${}^{23}\text{Na}$ " [1] — reduces, once optical pumping is required, to lithium and sodium, and lithium is the lighter, cluster-structured and ab initio-computable one (§1.2).

The source physics is established rather than speculative: polarized lithium beams have a forty-year history at low energy. The Heidelberg tandem delivered ${}^6\text{Li}^{3+}$ at vector polarization 0.57–0.65 in 1977 [30], and by 1987 its atomic-beam source reached $P_z, P_{zz} \geq 0.85$ and accelerated ${}^7\text{Li}$ at an alignment $P_{zz} = 0.69$ on target [31]; the optically pumped Florida State source delivered more than 150 nA of ${}^6\text{Li}^{3+}$ at $P_{yy} = -1.33 \pm 0.02$ with no polarization loss in the linac [32]; and polarized ${}^6\text{Li}$, ${}^7\text{Li}$ and ${}^{23}\text{Na}$ beams served two decades of nuclear-reaction studies [33]. New for the EIC are scale and transport: $\geq 10^{13}$ atoms/s, two to three orders of magnitude beyond these sources, and preservation of the polarization through charge breeding and acceleration beyond 100 GeV/u [51,1].

In the ring, ${}^6\text{Li}$ is the easy case. Its gyromagnetic anomaly, $G \approx -0.18$, is as small as the deuteron's (-0.14), so it crosses only 81 depolarizing resonances on the way to top energy, against 63 for the deuteron and 2457 for ${}^3\text{He}$ [28]; like the deuteron it is handled with a partial solenoid snake and tune jumps, full Siberian snakes being impractically long for so small a G [28,1], so the deuteron techniques [52] carry over. ${}^7\text{Li}$ ($G \approx +1.52$) crosses 573 resonances and needs partial snakes with jump quadrupoles [1,28]. The ring rigidity fixes the top energies at $275 \text{ GeV} \times Z/A$: 137.5 GeV/u for ${}^6\text{Li}$ and 117.9 GeV/u for ${}^7\text{Li}$. Because ${}^6\text{Li}$ has $A/Z = 2$ it runs at the same per-nucleon energy and magnet settings as the deuteron, so $e+d$ and $e+{}^6\text{Li}$ can be compared at identical kinematics.

1.2 · Why lithium as a target: the spin sits on a light cluster

The two stable lithium isotopes are the lightest nuclei beyond $A = 3$ in which a polarized light cluster is bound to an inert core: in ${}^6\text{Li}$ an α particle ($J = 0$, $T = 0$) and a deuteron-like cluster bound by 1.47 MeV against $\alpha+d$ breakup, in ${}^7\text{Li}$ an α and a triton bound by 2.47 MeV [36]. Variational Monte Carlo with realistic two- and three-nucleon forces makes this quantitative: the α -d overlap of the ${}^6\text{Li}$ ground state integrates to $N_{\alpha d} = 0.86$ (0.846 S-wave, 0.017 D-wave), the α -t overlap of ${}^7\text{Li}$ to $N_{\alpha t} = 1.00$ [17], and the no-core shell model with continuum gives an asymptotic D/S ratio of -0.027 for the $d+\alpha$ configuration [38]. In a fully polarized ${}^6\text{Li}$ the α contributes nothing and the two valence nucleons carry the spin: six-body VMC gives valence numbers 1.924 up and 1.076 down, a polarization of 0.85 [17,19]; three-body $\alpha+n+p$ models, in which $\alpha+d$ accounts for 60–70% of the wave function, give above 90% [18]; and E155, which used ${}^6\text{LiD}$ as its deuteron target, adopted 87% for the effective deuteron polarization [43]. ${}^6\text{Li}$ is therefore a polarized deuteron in the field of an α particle, whose tensor observables can be set against the free deuteron's to isolate the medium effect. The only other stable spin-1 ground state is ${}^{14}\text{N}$, whose spin is shared among p-shell nucleons, with no comparable cluster simplicity and beyond the reach of exact quantum Monte Carlo ($A \leq 12$) [40].

In ${}^7\text{Li}$ the spin is carried by the unpaired proton of the triton cluster, the neutrons being paired [39]. VMC gives proton numbers 1.934 up and 1.066 down and neutron numbers 1.981 and 2.019 [17], i.e. $P_p = 0.87$ and $P_n = -0.04$ (0.866 and -0.037 in the JLab proposal [45]); the shell model already gives $P_p = 13/15$, and the microscopic calculations confirm it and leave the neutrons almost unpolarized [19,14]. Polarized ${}^7\text{Li}$ is thus to the proton what polarized ${}^3\text{He}$ is to the neutron, with the property, decisive for the polarized EMC effect, that the proton sits in a nucleus of normal density. Cloët, Bentz and Thomas set the selection criteria — $|P_n| \ll |P_p|$ and light, because g_1^A is carried by the valence nucleons and suppressed relative to F_2^A by roughly $1/A$ — and their candidates are ${}^{11}\text{B}$, ${}^{15}\text{N}$, ${}^{27}\text{Al}$ and ${}^7\text{Li}$, "where the nuclear polarization is largely dominated by the valence proton" [14]; the gluon-spin study names ${}^7\text{Li}$ "the most promising case for experimental investigation" [16].

Two further properties matter. The cluster structure makes the nuclei computable: Green's-function Monte Carlo with AV18+IL7 reproduces both binding energies (-31.93 vs -31.99 MeV for ${}^6\text{Li}$, -39.15 vs -39.25 MeV for ${}^7\text{Li}$), the charge radii to within 3%, and, with two-body currents, the magnetic moments (0.835 vs $0.822 \mu_N$; 3.24 vs $3.256 \mu_N$) [40,39], so the nuclear corrections entering every extraction — effective polarizations, momentum distributions, spectral functions — carry controlled uncertainties rather than model spread. The quadrupole moment of ${}^6\text{Li}$ is the exception: the measured $Q = -0.0806 \text{ fm}^2$, against $+0.286 \text{ fm}^2$ for the free deuteron [36], arises from a delicate cancellation between the deuteron's own moment and the D-wave part of the α -d relative motion [19,37], and theory reproduces only its smallness. That smallness is an asset — ${}^6\text{Li}$ is nearly spherical, which bounds the deformation term of the coherent channel (§4.2) — while ${}^7\text{Li}$, whose p-shell triton orbit gives $Q = -4.00 \text{ fm}^2$ [36], is the deformed counterpart. The isotopes also share one source, one injector and one detector: the $\alpha+d$ versus $\alpha+t$ comparison changes only the cluster.

1.3 · Why polarization is indispensable

An unpolarized lithium beam would be one more light ion: its inclusive cross section is carried by F_1 and F_2 alone — for an unpolarized target the spin-1 cross section has exactly the spin- $1/2$ form [2] — and it would add an $A = 6, 7$ point to the EMC and shadowing curves that ${}^4\text{He}$ and ${}^{12}\text{C}$ already map. Every item of the program is instead a spin observable, and the two flagship items need the tensor moment specifically: b_1 is *defined* as the difference between the $m = 0$ and $m = \pm 1$ quark distributions, so averaging over m erases it, and Δ needs both a target spin of at least 1 and a transverse alignment axis to define an azimuth (§3). The polarized EMC effect is a statement about g_1 , and tagged DIS reads the spin of the cluster (§4).

Polarized lithium would be the first polarized ion at a collider with a nuclear core — the first place where the spin structure *of a nucleus*, rather than of a nucleon, is studied at collider kinematics — and tensor polarization, on which both b_1 and Δ depend, is the scarcer resource.

1.4 · Fixed target or collider?

Polarized lithium has existed for decades as a fixed target, but only in the solid state: dynamically polarized ${}^6\text{LiD}$ was the deuteron target of SLAC E155 and of COMPASS, ${}^7\text{LiH}$ has been polarized at Saclay (47% for ${}^7\text{Li}$, 56% for the protons) [44], and ${}^7\text{LiD}$ (with ${}^6\text{LiH}$ or NH_3 as the proton reference) is the material of the approved CLAS12 polarized-EMC experiment [43,42,45]. In these compounds the lithium polarizes along with the hydrogen isotope because all spin species share one spin temperature: COMPASS reached deuteron polarizations of +54%/−47%, whence $51 \pm 3\%$ for ${}^6\text{Li}$ and $92 \pm 4\%$ for the ${}^7\text{Li}$ admixture [42]; E155 measured ${}^6\text{Li}$ at 97% of the deuteron's [43]; in ${}^7\text{LiD}$ at 5 T the CLAS12 proposal expects 65–80% for ${}^7\text{Li}$ [45]. Three limits are structural. (i) Dilution: only $f \approx 0.35\text{--}0.45$ of the nucleons in ${}^6\text{LiD}$ are polarizable once the helium coolant is counted — better than 0.22 for ND_3 and 0.15 for NH_3 , far from the 1.0 of a stored beam [42,43]. (ii) Tensor polarization: in thermal equilibrium it follows from the vector one, $P_{zz} = 2 - \sqrt{4 - 3P_z^2}$, so $P_z = 0.5$ gives $P_{zz} = 0.20$; radio-frequency "hole burning" reaches 29–37% in deuterated butanol at the cost of the vector polarization, and the JLab b_1 experiment is conditionally approved on demonstrating 30% [8,46]. An atomic-beam source prepares the tensor moment directly: HERMES ran its gas target at $P_{zz} = +0.89$ and -1.65 , "a combination which is not possible for solid-state polarized targets" [5]. (iii) Spectator loss: a spectator α of 100–300 MeV/c carries only 1.3–12 MeV in the target rest frame and stops within 10–250 μm of lithium deuteride [49], inside cells 2–5 cm long at JLab and 30–60 cm at COMPASS, and a coherent ${}^6\text{Li}$ recoil at $|t| = 0.05 \text{ GeV}^2$ has only 4.5 MeV. The detectors that do tag slow spectators, BoNuS (protons from 70 MeV/c) and ALERT (α from 230 MeV/c), work with unpolarized gas at a few atmospheres [47,48]; polarized lithium outside dynamically polarized solids is essentially unknown (a 1975 target for a low-energy reaction study [50] is the one instance on record) and has never been used in lepton scattering.

The collider removes each limit at once. The beam is pure lithium, its polarization prepared atom by atom and measured in the ring; the spectator α or triton carries the beam momentum per nucleon into the Roman Pots, off-momentum detectors, B0 and zero-degree calorimeter at a few milliradians, with $\sim 10\%$ p_T resolution at 100 MeV/c [22]; the intact coherent recoil can be tagged down to $p_T \approx 0$ at the IR-8 secondary focus [27]; and $\sqrt{s_{NN}}$ up to 99 GeV for $e(18 \text{ GeV}) \times {}^6\text{Li}(137.5 \text{ GeV}/u)$ opens x down to $\sim 10^{-4}$ at $Q^2 \geq 1 \text{ GeV}^2$, where JLab at 11 GeV stops at $x \approx 0.06$ and HERMES stopped at 0.01 [45,5] — the region where gluons dominate, where HERMES saw its large b_1 , and where coherence holds. What stays with the fixed target is luminosity: $10^{35} \text{ cm}^{-2} \text{ s}^{-1}$ (CLAS12 at 10–15 nA, Hall C at 115 nA, the material annealed every 2–3 days) [45,8] against an electron–nucleon luminosity of $10^{33}\text{--}10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ at the EIC [29,25]. The high- x polarized EMC effect and the high- x b_1 of the deuteron are therefore rightly JLab measurements; the EIC takes the low- x and high- Q^2 regions, the tensor observables at full polarization, and everything that needs a detected spectator or recoil.

MEASUREMENT	AT A FIXED TARGET	WHAT ONLY THE COLLIDER ADDS
Polarized EMC effect, ${}^7\text{Li}$ (g_1^A)	Approved: JLab E12-14-001 (CLAS12, polarized ${}^7\text{LiD}$, $0.06 < x < 0.8$, $1 < Q^2 < 15 \text{ GeV}^2$, $P_z \approx 0.8$ assumed, 55 days) [45]	$x < 0.06$; the Q^2 evolution that carries the gluon-spin EMC effect [16]; an undiluted beam; α -tagging of the struck triton and its virtuality (IP6 Roman Pots)
Tensor b_1 , ${}^6\text{Li}$	Deuteron only: E12-13-011 (ND_3 , $0.16 < x < 0.49$, $P_{zz} \geq 0.3$ required) [8]; ${}^6\text{LiD}$ would give $P_{zz} \approx 0.2$ with the deuteron term to be subtracted, and no such measurement is proposed	x down to 10^{-3} , where HERMES found $b_1 \approx 0.1$; pure-tensor fills at $P_{zz} \approx 0.6\text{--}0.8$; spin axis along $\mathbf{q}$ [26]
Exotic glue Δ , ${}^6\text{Li}$ (${}^7\text{Li}$)	Letter of intent only: LOI12-16-006 (${}^{14}\text{N}$ in NH_3 , $x < 0.3$, 2–12% alignment, in-plane spectrometers — $\cos 2\phi$ "difficult with a fixed target") [13]	Transverse tensor polarization ≥ 0.6 with full 2π azimuthal acceptance; the gluon-dominated region $x \leq 0.1$ and a Q^2 lever arm
Spectator-tagged DIS (α , t)	Not feasible: spectators stop within 0.25 mm of a solid target; no polarized lithium gas target exists	Far-forward detection of α and t at beam velocity; pole extrapolation to the free cluster [20,22]

1.5 · What each observable requires of the beam, and what lithium alone provides

The reach below comes from the fast simulation and the polligen pseudo-experiments (§5) at the placeholder in-ring polarizations $P_z = 0.7$, $P_{zz} = 0.6$ and $P_e = 0.7$, with $10 \text{ fb}^{-1}/u$; transport of the source polarization to the ring is an open accelerator question. Every statistical error scales as $1/(P\sqrt{L})$, so halving P_{zz} costs a factor four in running time. Two machine facts shape the list: the stable spin direction in the arcs is vertical, so transverse polarization is native at the interaction point while longitudinal polarization needs spin rotators whose lithium settings are undefined [29,1]; and tensor fills need $m = 0$ bunches from the source radio-frequency transitions, whose survival through acceleration is not established [51]. The observables demand different spin configurations and share the luminosity, so each reach assumes the full $10 \text{ fb}^{-1}/u$ in its own configuration. (Semi-inclusive channels add the three leading-twist tensor TMDs of a spin-1 target [1], not projected here.)

OBSERVABLE	BEAM POLARIZATION REQUIRED	ELECTRONS	REACH AT $10 \text{ FB}^{-1}/U$	BINDING REQUIREMENT
g_1^A , polarized EMC (⁷ Li; ⁶ Li for the isoscalar pair)	Vector $P_z \geq 0.7$, longitudinal (spin rotators). A spin-3/2 spin-temperature fill at $P_z = 0.8$ also carries $P_{zz} \approx 0.55$ and $P_{zzz} \approx 0.11$; the octupole term enters the asymmetry numerator and is corrected through the fill model [45]	Polarized, $P_e \approx 0.7$, helicity flipped bunch by bunch	$\delta\Delta R \approx 2.6\text{--}4\%$ per x-bin at $x = 0.3\text{--}0.5$ (three energies combined); CBT-vs-TMT $\approx 1.5\sigma$ per bin, 5σ at $x \approx 0.5\text{--}0.7$ with $100 \text{ fb}^{-1}/u$	Polarimetry scale $\delta P_z/P_z \leq 3\%$; P_p from VMC [17]; Q^2 lever arm at fixed x (gluon-spin EMC)
b_1 via A_{zz} (⁶ Li)	Tensor P_{zz} , longitudinal or along q [26]; fills of different P_{zz} ("tensor thirds": $m = +1\text{--}, 0\text{--}, -1\text{-rich}$), i.e. $m = 0$ bunches from the source	Unpolarized (helicity-averaged)	$\delta A_{zz} \approx 1 \times 10^{-4}$ per x-bin (combined over Q^2 and the three energies) at $x < 0.05$ (7×10^{-5} at $P_{zz} = 0.8$) against scenario $ A_{zz} = (2\text{--}10) \times 10^{-3}$ (HERMES-like b_1 with the 0.29 dilution) or ten times smaller (convolution): both $\geq 10\sigma$ per bin. At $x > 0.3$, $\delta A_{zz} \approx 10^{-3}$ against a signal $\leq 2 \times 10^{-3}$ — JLab's domain	Relative luminosity between fills at the 10^{-4} level (a 10^{-4} offset biases A_{zz} by 1×10^{-4} ; the luminosity-corrected estimator removes it exactly); tensor polarimetry, for which no method is established [1]
Δ via $\cos 2\phi$ (⁶ Li; ⁷ Li eligible)	Tensor P_{zz} with the alignment axis transverse (the native vertical direction); fills of opposite P_{zz} sign flip the modulation — the decisive systematic check	Unpolarized	$\delta A = (1.5\text{--}4.5) \times 10^{-4}$ per sweet-spot bin ($x \approx 0.02\text{--}0.14$, $Q^2 = 1.1\text{--}14 \text{ GeV}^2$) against moment-constrained amplitudes of $(0.7\text{--}1.2) \times 10^{-2}$, $\geq 21\sigma$; a 3×10^{-3} flat amplitude stays $3\text{--}10\sigma$ per bin even at $P_{zz} = 0.3$. Reconstructed (two-state ratio, hadronic y from a PYTHIA 8 final state): $\delta \hat{A} = (1.0\text{--}3.0) \times 10^{-4}$, bin purities 0.40–0.73, amplitude 4–21% below the true-bin value	Spin-state-sorted ratio of $m = \pm 1\text{-rich}$ and $m = 0\text{-rich}$ bunches, alternating bunch by bunch (§5); hadronic-method y at the $y \approx 0.01$ sweet spots, where the Σ -method $\delta y/y$ with 50 MeV of calorimeter noise is 0.55 / 0.28 / 0.50 / 0.15 at the mid configuration, 0.28 / 0.21 / 0.24 / 0.11 at low and 0.74 / 0.34 / 0.69 / 0.18 at top — the $x \approx 0.1$ bins therefore belong to low-energy running — and where the noise floor decides the answer (0.20 / 0.32 / 0.54 / 1.01 at $y = 0.005$ for 0 / 25 / 50 / 100 MeV, on a hadronic $E - p_z$ sum of 0.2–0.5 GeV that no published ePIC study covers); $\delta P_{zz}/P_{zz} \approx 3\%$ scale; tensor

				radiative corrections not yet calculated
Spectator-tagged DIS	${}^7\text{Li}$ α -tag: any polarization for tagged F_2 ; vector P_z with polarized electrons for tagged g_1 of the bound proton. ${}^6\text{Li}$ α -tag: tensor P_{zz} for the order-unity tagged tensor asymmetries at $p_s \approx 300$ MeV/c [41]	As for the inclusive observable	${}^7\text{Li}$ $\alpha \rightarrow$ Roman Pots ($R = 0.856$), 96–99% acceptance, optics-robust; ${}^6\text{Li}$ α beam-blind at IP6 (1.85% / 1.51% at 137.5 GeV/u in the high-acceptance / high-divergence optics and 13.5% at 99.5 GeV/u, with the near-beam cut applied as an angle and the beam boost from the physical nuclear mass) $\rightarrow$ IR-8 secondary focus	Spectator p_T resolution $\sim 10\%$ at 100 MeV/c [22]; charge identification of α against $A/Z = 2$ fragments in the Roman Pots; pole extrapolation in p_T [20]. The ${}^6\text{Li}$ α tag is entirely a p_T -tail measurement, and the cluster model undershoots the official BeAGLE e+d sample by $2.5\times / 6.9\times / 28\times$ at $p_T > 0.2 / 0.3 / 0.45$ GeV, so its model uncertainty is one-sided upward
Coherent intact- ${}^6\text{Li}$ $\cos 2\phi$	Tensor $P_{zz} \geq 0.6$, transverse	Unpolarized	$x \lesssim 0.01$, $ t < 0.2$ GeV ² ; 1.1×10^7 tagged events (13.5% tag acceptance for the constant 0.20 GeV cut in the high-acceptance optics; $f_0 = 0.04$ scenario), best-bin $\delta A = 1.9\times 10^{-3}$ (5σ floor at a 1.0% modulation); deformation-anchored ($a_z = 0.036$ (band 0.018–0.059; the anchor's convention makes the $\cos 2\beta$ coefficient $2a_z$); a flat gluonic 3×10^{-3} – 10^{-2} gives 1.6– 5σ , or 5– 16σ with $100\text{ fb}^{-1}/\text{u}$. Reconstructed with the angular envelope (0.22 GeV at 99.5 GeV/u) and a slot-shaped cutout: 2.7×10^6 tagged events, $a_t = 0.117 \pm 0.007$ and $a_e = 0.010 \pm 0.001$ in year 1 (background-free by assumption). With the measured pot aperture (§4.2) only the low-energy configuration survives: acceptance 1.42% against the envelope's 37.7%, 3.1×10^5 tagged events, two of the four $ t $ bins, δa_t worse by 6–34 $\times$, $a_e = 0.0073 \pm 0.0045$ and 0.0091 ± 0.0045 against an injected 0.0100	High-acceptance optics (67% / 9% / 10^{-8} at 40.8 / 99.5 / 137.5 GeV/u for the angular envelope; $1.4\times 10^{-2} / 5\times 10^{-5} / 1.9\times 10^{-17}$ for the measured pot aperture, tighter than the envelope) or IR-8; Z-identification against α +d (both at beam rigidity); B0 calorimeter veto of the 3.56 MeV de-excitation γ

Species by species: ${}^6\text{Li}$ is the only $A > 2$ nucleus with $J \geq 1$ the EIC could polarize — b_1 and Δ do not exist for protons or ${}^3\text{He}$ at all — and the lattice indication that the deuteron's exotic glue is suppressed [11] makes a bound nucleus the likelier place for a discovery [1]. ${}^7\text{Li}$ is the only bound polarized proton within reach of any collider, and the only light-ion beam whose spectator is a spin-0, tightly bound nucleus landing in the Roman-Pot window today; ${}^6\text{Li}$ and the deuteron supply the isoscalar companions, so the three together separate the u- and d-quark spin modification — an inference of this program, not yet in the literature.

OBSERVABLE	P	D	${}^3\text{He}$	${}^6\text{Li}$	${}^7\text{Li}$
g_1^A , polarized EMC	free reference	isoscalar pair, weakly bound; D-wave depolarization of a few per cent [20]	neutron ($P_n = 0.86$ – 0.88) at $A = 3$	isoscalar pair, $P = 0.85$, in the α field	the bound polarized proton , $P_p = 0.87$, at normal density
b_1	—	HERMES; EIC deuterons a later addition	—	the only $A > 2$ case ; ${}^6\text{Li}$ – d isolates the α field	rank-2 analog, no published formalism
Δ (exotic glue)	— ($J = \frac{1}{2}$)	allowed; lattice suggests suppression [11]	—	allowed, bound ; spin-1 like the ϕ meson of the lattice signal [10]	allowed; $J = 3/2$ as in the bag estimate [9]

Spectator tagging	—	p, n spectators (off-momentum detectors, ZDC)	p spectators into the Roman Pots; double tag [21]	α beam-blind $\rightarrow$ IR-8; tagged tensor asymmetries $O(1)$	α into the Roman Pots now ; struck A = 3 cluster
Coherent tensor $\cos 2\phi$	—	deformation term dominant [23]	—	deformation null ($ Q $ 3.5 $\times$ smaller than d, opposite sign)	deformed reference

What an unpolarized lithium beam already gives. Three things, none of them the driver. Nuclear parton distributions at A = 6 and 7: the high-x EMC ratios of ${}^6\text{Li}$ and ${}^7\text{Li}$ are in the target list of JLab E12-10-008 [54], motivated by the E03-103 finding that the ${}^9\text{Be}$ EMC effect follows the local density of its α clusters rather than its average density [53] — a picture predicting an effect for ${}^6\text{Li}$ and ${}^7\text{Li}$ well below the other models [54] — with the EIC adding the low-x shadowing region. Cluster virtuality: the α -tagged F_2 of the triton (${}^7\text{Li}$) or deuteron (${}^6\text{Li}$) cluster versus spectator momentum measures the α -t and α -d momentum distributions VMC predicts ($N_{\alpha t} = 1.00$ in a relative p-wave, $N_{\alpha d} = 0.86$) [17], the collider counterpart of the JLab tagged-EMC program [48]. And coherent diffraction on an α -clustered nucleus, whose t-distribution images the gluon density of the cluster, as studied for ${}^7\text{Li}$ at IR-8 [27]. Polarization turns the same structure into a tool: b_1 , Δ , g_1^A and the tagged tensor asymmetries all vanish or are undefined without it.

2 • Deep inelastic scattering

An electron with four-momentum k scatters from a nucleus of momentum P by exchanging a virtual photon $q = k - k'$. The kinematic invariants, in the per-nucleon convention used throughout the program (energies and structure functions per nucleon, $x \leq 1$), are

$$Q^2 = -q^2, \quad x = \frac{Q^2}{2P \cdot q}, \quad y = \frac{P \cdot q}{P \cdot k}, \quad Q^2 = xys, \quad W^2 = M^2 + Q^2 \frac{1-x}{x}$$

where x is the momentum fraction of the struck parton, y the energy-transfer fraction, and W the invariant mass of the hadronic final state. The unpolarized cross section is carried by two structure functions,

$$\frac{d^2\sigma}{dx dQ^2} = \frac{4\pi\alpha^2}{xQ^4} \left[\left(1 - y + \frac{y^2}{2}\right) F_2(x, Q^2) - \frac{y^2}{2} F_L(x, Q^2) \right], \quad F_1 = \frac{F_2}{2x(1+R)}$$

with $R = \sigma_L/\sigma_T$, taken throughout from the SLAC/E143 R1998 world fit [55]. In the parton model, $F_2 = \sum_q e_q^2 x[q(x) + \bar{q}(x)]$: the cross section measures quark distributions weighted by squared charge. For a nucleus we use F_2^A/A , so rates are quoted against luminosity per nucleon. Polarization introduces additional structure functions beyond F_1 and F_2 ; their number and character depend on the target spin.

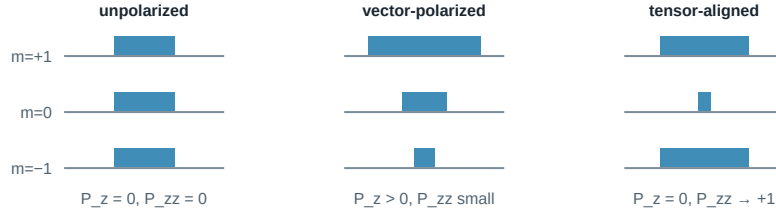
3 • Spin-1 targets

A spin- $1/2$ target contributes two spin structure functions, g_1 and g_2 . The principal observable is g_1 , the helicity-weighted quark distribution, measured through the double-spin asymmetry $A_{\parallel} \approx D \cdot g_1 / F_1$ (D is a kinematic depolarization factor) with both beams longitudinally polarized and the electron helicity reversed bunch by bunch.

A spin-1 target carries additional structure. Its spin state along a chosen axis is specified by three populations p_+ , p_0 , p_- of the projections $m = +1, 0, -1$, and two moments characterize the fill:

$$P_z = p_+ - p_- \in [-1, 1], \quad P_{zz} = p_+ + p_- - 2p_0 = 1 - 3p_0 \in [-2, 1]$$

P_z is the familiar vector polarization. P_{zz} , the *tensor* or alignment polarization, is specific to spin ≥ 1 : it distinguishes spin along $\pm z$ from spin in the transverse plane ($m = 0$) and is nonzero even when equal numbers point up and down. An unpolarized beam has ($1/3, 1/3, 1/3$) and both moments vanish; a pure $m = 0$ beam has $P_z = 0$ but $P_{zz} = -2$.



Hoodbhoy, Jaffe, and Manohar showed [2] that a spin-1 target has, beyond F_1 , F_2 , g_1 , g_2 , four tensor-polarized structure functions b_1 – b_4 , and Jaffe and Manohar identified one more, the double-helicity-flip function Δ [3]. At leading twist the tensor sector reduces to b_1 (with $b_2 = 2x b_1$ as its Callan–Gross partner) and Δ . For unpolarized electrons on a target whose spin axis makes angle θ_m with the beam, the cross section for each pure m state is [2,3]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1) \text{ for } m = (+1, 0, -1)$$

Sign convention. The b_1 , b_2 terms are written in the transcription used throughout the program. In the Hoodbhoy–Jaffe–Manohar convention used by HERMES and by Cosyn et al. [26] (their Eq. 27, $A_{zz} = -(2/3) b_1/F_1$ for the axis along the photon) the tensor-rate terms carry the opposite sign, so the $m = 0$ state has the larger rate for $b_1 > 0$. Only the sign of A_{zz} relative to b_1 is affected; the magnitudes, the errors and the Δ term are not.

The weights (1, -2, 1) organize all tensor observables: the $m = \pm 1$ states enter together and against $m = 0$, precisely the combination measured by P_{zz} , and averaging over a fill replaces every c_m by P_{zz} . Two handles follow: with the spin axis *longitudinal*, the rate difference between fills gives $A_{zz} \propto b_1$; with it *transverse* ($\theta_m = 90^\circ$), the ϕ distribution of the scattered electron acquires a $\cos 2\phi$ modulation proportional to Δ . Both require tensor polarization; neither requires polarized electrons.

^7Li , with spin 3/2, has four populations and three moments (vector, tensor/quadrupole, octupole). Its rank-0 and rank-1 sectors are exact analogs of the spin-1/2 case and host the polarized EMC measurement; its rank-2 sector is the formal analog of the spin-1 tensor sector, for which no structure-function basis has been published.

4 • The measurement program

4.1 • The tensor structure function b_1

In parton language, b_1 compares quark distributions in an aligned versus a transverse nucleus [2]:

$$b_1(x) = \frac{1}{2} \sum_q e_q^2 \left[q^{(0)}(x) - \frac{q^{(+1)}(x) + q^{(-1)}(x)}{2} \right], \quad \int_0^1 b_1(x) dx = 0 \quad (\text{Close – Kumano, no tensor sea})$$

where $q^{(m)}$ is the quark distribution in a target with spin projection m . For free spin-1/2 nucleons, or a pure S-wave bound state, all m states are equivalent and $b_1 = 0$, so a nonzero b_1 measures directly how binding modifies the partonic content: D-wave admixture, pion exchange, hidden-color components, or a tensor-polarized sea (which would violate the Close–Kumano sum rule [4]).

The only existing measurement is HERMES on the deuteron [5]: A_{zz} up to 2×10^{-2} over $0.01 < x < 0.45$ and $b_1(x = 0.012) = 0.112 \pm 0.055 \pm 0.028$, unexpectedly large at low x , with a zero crossing near $x \approx 0.2$ and a sum-rule integral 2σ from zero. Convolution calculations predict $|b_1| < 10^{-3}$ at $x \gtrsim 0.2$, an order of magnitude below the data [6], whereas a model with pions and a hidden-color probability of only 0.15% reproduces it [7]. JLab E12-13-011 will remeasure $0.16 < x < 0.49$ with a solid ND_3 target [8]; the EIC adds the low- x region and the Q^2 lever arm.

Measurement procedure. The beam is delivered in fills of different tensor polarization ("tensor thirds"); with the estimator $(N_+ + N_- - 2N_0)$ over matched luminosities, all m -independent physics cancels (§5). At collider kinematics a clean extraction favours orienting the spin axis along the momentum transfer $\mathbf{q}$ rather than along the beam [26].

4.2 · $\Delta(x, Q^2)$: exotic glue

The $\cos 2\phi$ term corresponds to the virtual photon flipping its helicity by two units between absorption and emission in the forward Compton amplitude. A spin- $\frac{1}{2}$ particle cannot absorb two units of helicity, so neither quarks nor bound nucleons generate Δ at leading twist; gluons can. Δ is what the gluon transversity distribution — an interference of gluon helicity states existing only for $J \geq 1$ — generates at order α_s through a photon–gluon convolution [3,9], so a nonzero Δ in a nucleus is direct evidence for glue not inherited from individual nucleons, hence "exotic glue". It has never been measured; the one prior experimental document is a JLab letter of intent [13].

The measurement uses unpolarized electrons on a tensor-polarized nucleus whose alignment axis lies in the transverse plane, and fits the azimuthal distribution about that axis. Averaged over the fill, the amplitude and its inversion back to Δ are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2$$

Writing Δ as Δ/F_1 and $F_2 = 2x(1+R)F_1$ removes both structure functions from that ratio and leaves R as the only one that reaches the physics result — F_2 cancels exactly. Taking R from the published R1998 fit [55] rather than a simplified form moves Δ/F_1 by +16.6 / +18.0 / +4.7 / -4.4% at the four sweet spots and roughly halves the luminosity the fast simulation needs for 5σ at $\Delta/F_1 = 10^{-3}$: 67.5 / 65.8 / 155.1 fb $^{-1}$ /u at the low, mid and top beam configurations, against 135.3 / 131.3 / 274.6 with the simplified form.

The expected size spans per-mille to percent, and the only absolute normalizations available are moment estimates: a bag-model calculation by Sather and Schmidt for the spin-3/2 Δ^{++} baryon, $\int x \Delta dx = -0.012 \alpha_s$ [9], and lattice-QCD calculations by Detmold and Shanahan of gluon transversity in the ϕ meson (clearly nonzero, at an unphysical pion mass) [10], with NPLQCD finding the analogous matrix element roughly ten times suppressed in the deuteron [11]. Percent-level asymmetries appear only in proton–deuteron Drell–Yan under the maximal assumption $\Delta_T g = \Delta g$, which its authors call optimistic [12]. No calculation exists for ${}^6\text{Li}$ or any $A > 2$ nucleus, and the literature brackets the nuclear Δ both ways: suppressed, as in the lattice deuteron, or enhanced by binding in analogy with the EMC effect [1]. The projection report transfers the bag moment to ${}^6\text{Li}$ as a labelled scenario; one EIC year measures the amplitude at $\geq 20\sigma$ per bin.

The coherent channel. The same modulation can be measured in coherent diffraction, $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$, where the nucleus stays in its ground state and recoils intact. There the modulation mixes two mechanisms: the nuclear *shape*, a deformed gluon density modulating the diffractive slope (computed for polarized deuterons in the only existing calculation of this kind [23]), and exotic glue proper, which persists at small x where it connects to the elliptic-gluon program [24]. ${}^6\text{Li}$ discriminates between them because its quadrupole moment is anomalously small ($Q = -0.0806 \text{ fm}^2$, roughly 50 times smaller than ${}^7\text{Li}$'s): the shape term is bounded and linear in t , so a flat-in- t modulation cannot be a shape effect. Detecting the intact recoil is itself nontrivial: the nucleus keeps the beam rigidity and emerges within ~ 2 mrad of the beam, visible only in the Roman-Pot transverse-momentum tail at IP6 or at the secondary focus of a second interaction region [25,27]. The pot aperture for an intact ${}^6\text{Li}$, measured by tracking one through the ePIC far-forward geometry, opens *horizontally* — $|\theta_x| \gtrsim 2.0 / 1.35 / 1.03 \text{ mrad}$ in the $5 \times 41 / 10 \times 100 / 18 \times 275$ optics against $|\theta_y| \gtrsim 1.8\text{--}3 \text{ mrad}$ — because the beamline images an interaction-point angle onto the pot plane with a horizontal lever of 30.6 m against a vertical few metres (one event per scan point, 30° azimuthal steps, no beam envelope, September-2024 `epic-main` geometry). That aspect is the opposite of the slot the chain assumed, so the acceptance-induced $\langle \cos 2\beta \rangle$ changes sign, $+0.426 \rightarrow -0.772$, and only the low-energy configuration keeps accepted recoils.

4.3 · The polarized EMC effect: g_1 of a bound proton

The EMC effect — the $\sim 10\text{--}20\%$ suppression of F_2 per nucleon in a nucleus at $0.3 \lesssim x \lesssim 0.7$ — has been measured for forty years without a settled explanation. Its spin analog has never been measured at all: does the *spin* structure of a bound nucleon change more, less, or the same as its momentum structure? The observable is the polarized EMC ratio

$$\Delta R_A(x) = \frac{g_1^A(x)}{P_p g_1^p(x) + P_n g_1^n(x)}$$

where P_p, P_n are the effective nucleon polarizations from nuclear structure. ${}^7\text{Li}$ is well suited: $P_p = +0.87$ against $P_n = -0.04$ (§1.2) [17,45], so it is to a good approximation one polarized proton plus six spectators, with computable corrections. Two classes of model differ sharply: Cloët–Bentz–Thomas (mean-field, NJL) obtain a polarized EMC effect about twice the unpolarized one [14], Tronchin–Matevosyan–Thomas equal effects [15]. A measurement at the 5% level discriminates between them, and the two map onto the competing mean-field and short-range-correlation pictures of the EMC effect itself. Unique to the EIC is the Q^2 evolution of g_1 , driven by the polarized gluon distribution; the predicted *gluon-spin* EMC effect is larger still, with ${}^7\text{Li}$ the most promising case [16], and the collider measures it free of target-fragment absorption.

Both beams are polarized longitudinally and the electron helicity flips bunch by bunch; the error scales as $1/(P_e P_z \sqrt{N})$, so the measurement uses the vector polarization the source delivers most readily.

4.4 • Spectator-tagged DIS

The cluster description of ${}^6\text{Li}$ is quantitative (§1.2): the α is inert and the deuteron-like pair carries the nuclear spin at a polarization of 0.85–0.9 in six- and three-body calculations [17,18,19], the program using E155's 0.87 [43]. Detecting the α spectator, which carries beam velocity into the far-forward spectrometers, identifies the struck object as the deuteron cluster, so the ${}^6\text{Li}$ beam acts as a source of polarized deuterons at controlled virtuality. The framework is spectator tagging (Cosyn–Weiss [20]): the spectator momentum is measured and extrapolated in its transverse component to the free-cluster pole, recovering free-particle structure with binding effects removed by construction rather than modelled. Tagged *tensor* asymmetries reach $O(1)$ at spectator momenta of ~ 300 MeV/c, where the D-wave dominates [41], far larger than the inclusive asymmetries. The benchmark is the $e+{}^3\text{He}$ double-tagging study of Frišić et al. [21], where tagging improved on effective-polarization unfolding by a factor 2–3, and 10 at low x .

Far-forward routing determines where each isotope's tag fragment is detected, and it follows from magnetic rigidity alone: a beam-velocity fragment bends in the hadron optics according to the ratio of mass-to-charge ratios $R = (m/Z)_{\text{frag}}/(m/Z)_{\text{beam}}$, of which $(A/Z)_{\text{frag}}/(A/Z)_{\text{beam}}$ is the $A \cdot M_u$ approximation — good to 5×10^{-3} , but the α is more bound per nucleon than the lithium it leaves, so it sits below beam rigidity rather than exactly on it.

BEAM	TAG FRAGMENT	R	DESTINATION AT IP6
${}^7\text{Li}$ ($A/Z = 7/3$)	α	0.85571	Roman Pots main window ($R \in 0.60\text{--}0.95$) — works now
${}^7\text{Li}$	t	1.28971	over-rigid, no coverage
${}^6\text{Li}$ ($A/Z = 2$)	α / d	0.99813 / 1.00452	within 5×10^{-3} of beam rigidity: inside the beam envelope except the p_T tail
${}^6\text{Li}$	p / n	0.50251 / –	Off-Momentum Detectors / ZDC (veto handles)

The tagging program therefore begins with ${}^7\text{Li}$ at IP6, while ${}^6\text{Li}$ α -tagging — and the coherent intact- ${}^6\text{Li}$ channel, which faces the same rigidity problem — motivates the secondary focus of the second interaction region [27].

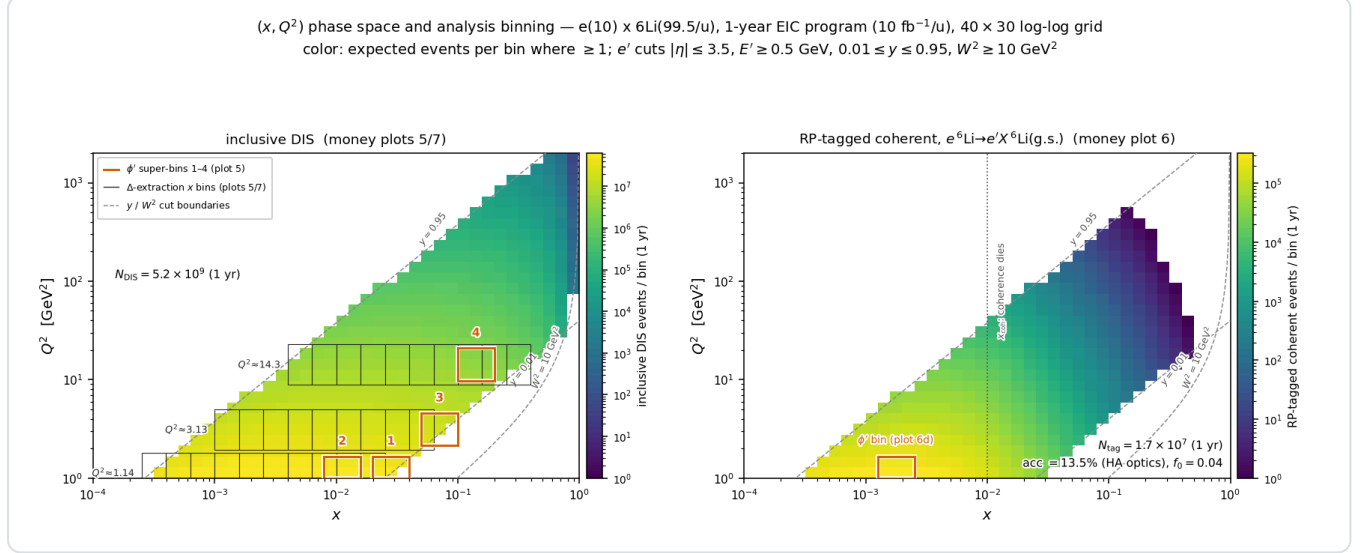
5 • Rates and statistical precision

All projections use luminosity per nucleon: $10 \text{ fb}^{-1}/u$ — 30 weeks at $10^{33} \text{ cm}^{-2} \text{ s}^{-1}$ with 60% efficiency in the Yellow Report's accounting [25], "one EIC year" — and $100 \text{ fb}^{-1}/u$ for the ten-year program, at the polarizations of §1.5. The statistical uncertainties behind every money plot are

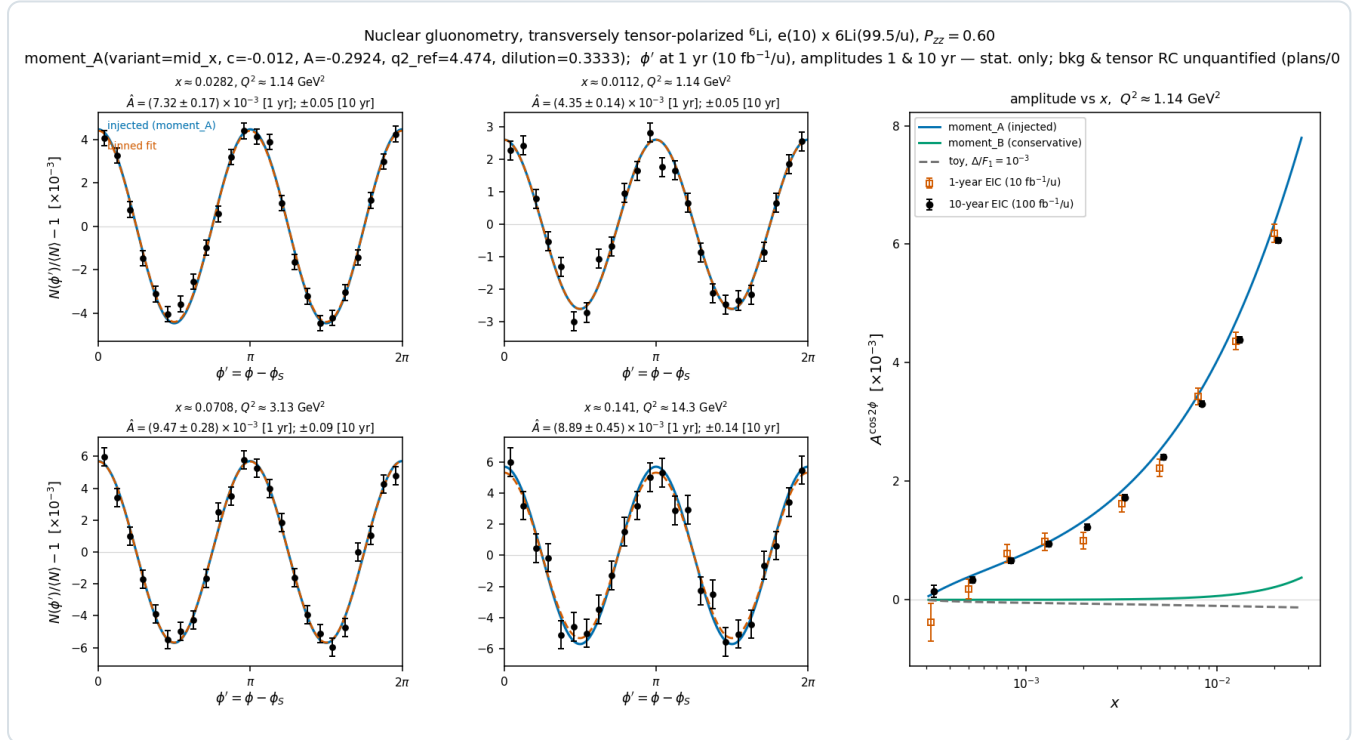
$$\delta A_{\parallel} = \frac{1}{P_e P_z \sqrt{N}}, \quad \delta A_{zz} = \frac{\sqrt{2/N}}{P_{zz}}, \quad \delta A^{\cos 2\phi} = \frac{\sqrt{2/N}}{P_{zz}} \text{ (single state)}, \quad \delta \hat{A} = \frac{2}{P_+ - P_0} \sqrt{\frac{2}{N}} \text{ (two states)}$$

The factor $\sqrt{2}$ in the tensor uncertainties follows from $\langle \cos^2 2\phi \rangle = \frac{1}{2}$ (equivalently, from the three-fill estimator variance); the polarization sits in the denominator because the fill dilutes the physical asymmetry. These formulas, and their behaviour under realistic ϕ -acceptance holes (where a binned fit replaces the naive moment), are validated end to end by polligen pseudo-experiments. Tensor observables have a structural advantage when the $m = \pm 1$ -rich and $m = 0$ -rich states are

measured with the same detector and compared bin by bin: in their spin-state ratio the ϕ acceptance cancels and relative-luminosity errors enter only as a constant, where a single-state fit instead measures the detector's own $\cos 2\phi$ acceptance harmonic divided by P_{zz} ; and the two-state pattern $P_+ = +0.6$, $P_0 = -1.2$ is $1.5\times$ more precise than a single state at the same source purity. The cancellation requires bunch-by-bunch alternation: a 10^{-3} difference of the acceptance harmonic between the states fakes half the signal (reconstruction-chain report §3). The polarimetry scale and the tensor radiative corrections (§1.5) do not cancel.



Phase-space coverage. Expected events per (x, Q^2) bin in one EIC year at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV}/u)$. Left: inclusive DIS (4.6×10^9 accepted events) with the bins of the $\cos 2\phi$ analysis overlaid; the measurement is limited by polarization, acceptance and systematics, not by statistics. Right: the coherent intact- ${}^6\text{Li}$ channel (1.1×10^7 tagged events at the constant 0.20 GeV cut, 2.7×10^6 with the angular envelope and slot cutout; with the measured pot aperture the channel survives only at the low-energy configuration, at 3.1×10^5 — §4.2), confined to $x \leq 0.03$ because coherence requires the photon to resolve the whole nucleus, which fails once $x \geq 0.01$.



Projected extraction. $\cos 2\phi'$ pseudo-data (one EIC year, $P_{zz} = 0.6$) in the four highest-significance (x, Q^2) bins, with the injected moment-constrained Δ model and the binned fit; right, the extracted amplitude versus x against alternative model readings. The precision distinguishes between the models rather than only from zero. The reconstructed-level version (money plot 5R, reconstruction-chain report §7) uses the spin-state ratio in reconstructed bins.

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Appendix A · Modification history

The main text states current content; this appendix records what changed and when.

DATE	REVISION
2026-08-17	First version: the DIS and spin-1 formalism from first principles, and the four flagship measurements (b_1 , Δ , the polarized EMC effect, spectator-tagged DIS) with their reach.
2026-08-21	§1 rewritten against primary sources — why lithium as a source and as a target, why polarization is indispensable, fixed target versus collider — and §1.5 added.
2026-08-25	References verified against the local copies in refs/: [28] corrected to E. Hamwi and G. H. Hoffstaetter, <i>Polarization transmission in the Electron-Ion Collider's Hadron Storage Ring</i> (J. Devlin is acknowledged, not an author; the title previously cited does not exist); [41] corrected to <i>Semi-inclusive DIS on a polarized spin-1 target</i> , Parts I/II.
2026-08-25	§1.5 table: the spin-state ratio and bunch-by-bunch alternation as the Δ requirement, hadronic-method y , the reconstructed-level 5R and 6R numbers; the b_1 , b_2 sign convention in §3; the two-state error formula and the systematics of §5. Modification history moved to this appendix.
2026-08-26	Development run 8 folded in: the PYTHIA 8 hadronic final state with its resolutions, purities and reconstructed dilution; the Roman-Pot aperture for an intact ^6Li measured in the ePIC geometry and carried through the coherent channel; R from the published R1998 fit [55]; physical nuclear masses in the far-forward routing; the BeAGLE-calibrated one-sided p_T -tail uncertainty. Prose cut throughout — process narration, work planning, superseded comparisons, repetition — and the overview table of §1 dropped as a duplicate of the §1.5 table.

Educational companion to [reports/cos2phi_money_plots_report.html](#) (projections and caveats), [reports/reconstruction_chain_report.html](#) (the measurement chain and the reconstructed-level results) and [plans/01_findings_physics_case.md](#) (sourced physics case). Open items of the simulation chain are in [plans/08_simulation_chain_completion.md](#) and the external inputs in [plans/04_open_questions.md](#). Generated by [reports/build_report.py](#); display math typeset at build time (matplotlib mathtext, STIX), figures from the evgen scripts. Projections use scenario structure functions with explicit bands.